

AI-Based Port Operations & Container Congestion Management

Problem Statement

Modern container ports face several operational problems due to unpredictable vessel arrivals, yard congestion, inefficient truck-gate flow, and manual resource allocation. These issues increase waiting time, turnaround time, and operational costs.

Major Problems

- **Unpredictable Vessel ETA:** Weather conditions and previous delays can change the actual arrival time of vessels, causing sudden berth bottlenecks.
- **Yard Congestion:** Uneven distribution and high container volumes can overload particular yard blocks and reduce handling efficiency.
- **Truck-Gate Congestion:** Uncoordinated truck appointments can create heavy traffic and long waiting times during peak hours.
- **Manual Berth Scheduling:** Manual allocation of berths and cranes may lead to inefficient resource utilization and vessel waiting.
- **High Turnaround Time:** Poor coordination between vessels, yard operations, trucks, and available resources increases the total time required for port operations.